

must start with zero.

- 16** Two coherent light waves that are same plane polarized of equal frequency and intensity superpose at a point that is equal distance from the two sources. The intensity of light at that point is twice the intensity of light due to either wave alone.

What is a possible phase difference between the two waves at that point?

A $\frac{\pi}{2}$

B π

C 2π

D 3π

Ans: A

Try drawing two sinusoidal waves that have a phase difference of $\frac{\pi}{2}$. You can do this by drawing a sine curve and a cosine curve, each of amplitude A , on the same grid. You will see that the resultant wave has an amplitude of $1.414A$ (when horizontal axis is $\frac{\pi}{4}$ rad). When $1.414A$ is squared and multiplied with constant k , it will give you $2I$ of intensity.

- 17** The stationary wave shown below is the result of the superposition of two identical waves